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Recent tendencies of extreme heat events in Italy

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Abstract

Over the last 20 years, several record-breaking events linked to high temperatures have enhanced the attention over extreme heat phenomena, especially during summer. Both 2022 and 2023 have been exceptional years in this regard. Heat waves, in particular, are a major natural hazard for the most fragile groups of the population. In order to address the points of impact, adaptation and future predictions, a quantitative analysis of all their specific aspects is necessary. These include the monitoring of their changes over time.

This study consists of a comprehensive analysis at national level of several temperature-extremes indices over Italy, including those specifically related to heat waves. The dataset is based upon a set of high quality daily maximum and minimum temperature records, from more than 250 ground-based weather stations. The time frame considered ranges from 1991 to 2020, providing information about the present Italian climate.

Two analysis are reported. One is targeted to find local features and is based on single stations data. The second one looks for a global signal related to heat extremes. The double level of the information (local and global) can be very useful in the comparison of results. The two methods are in agreement in describing an increasing presence of heat extremes, in terms of overall frequency, duration and intensity. Some areas appear as hot spots for the observed tendencies.

Keywords: temperature, heat extremes, heat waves

1 Introduction

The year 2022 has seen several all-time records concerning extreme weather conditions in Europe (2023). The scarcity of rain, as well as record air and sea surface temperatures, have put Europe into severe conditions regarding agriculture and health risks for the population European Commission, Joint Research Centre et al. (2022). From a wider perspective, recent decades have seen over Europe a general increase of heat extremes, as well as heat stress-related indices Lorenz et al. (2019). Moreover, compound events of heat and drought have been seen to particularly affect several hot spots of the continent, including Central and Southern Europe Bezak and Mikoš (2020). Along the latest ~20 years, the worst seasons regarding heat extremes can be found: 2003 for western central and southern areas García-Herrera et al. (2010), 2010 for eastern ones Barriopedro et al. (2011), 2018 over wide areas of the continent Gouveia et al. (2022) and 2022 for south-west and center again. All these events support the evidence that the frequency and intensity of extremely hot periods in Europe is increasing Lhotka and Kyselý (2022), coherently with the latest predictions from the IPCC Seneviratne et al. (2021).

Italy, in particular, has experienced during 2022 its second all-time hottest summer after 2003. Temperature above normal values have also been recorded in autumn and winter on average over the whole country, marking 2022 as the hottest year from 1961 (2023). Like Europe, the two latest decades have seen a general tendency towards extreme temperature records. From 2000, only four years show an overall negative mean temperature anomaly, with respect to the 1991-2020 climatological reference (2023). The next decades are finally not foreseen to see a change in this worrisome trend, since Europe could experience end-of-century climate scenarios for heat and drought much sooner than expected Suarez-Gutierrez et al. (2023).

Strongly related to high temperatures, heat waves are among the most dangerous of natural hazards, having high impact on society in terms of heat-related deaths. A recent study estimated more than 60.000 heat-related deaths in Europe between May 30th and September 4th, 2022, with the highest mortality value and rate in Italy Ballester et al. (2023). 2023 summer has also recorded a series of heat waves hitting Europe and the whole year ranked globally 1st among the warmest years on record. Those extreme events pose a serious hazard with respect to the environment and the most vulnerable groups of the population.

In order to fight against these severe weather conditions, which are evolving towards an increased exposure, today many countries have found a common strategy, which is based not only on mitigation of climate change, but also on adaptation to the new scenarios (2021). At this purpose, a fundamental key is the quantitative addressing of all the aspects that compose the climate change, such as the change in frequency, duration and intensity of certain phenomena.

There is no common definition of heatwaves Perkins and Alexander (2013), but different metrics have been employed to identify heatwaves and a variety of indices have been developed to describe their features. Using different approaches, a lot of studies have shown a worsening of heatwave features both at global and regional scale Perkins et al. (2012); Kueh et al. (2017); Perkins-Kirkpatrick and Lewis (2020); Morsy and

El Afandi (2021); Díaz-Poso et al. (2023); Galanaki et al. (2023). Regarding the Italian area, warm spells have been observed to be correlated with those over the whole Mediterranean basin Baldi et al. (2006). In this same area, from the 1970s to early 2000s an increase in the occurrence frequency of heat wave days has been observed, together with a significant decrease of cold days.

So far, dedicated analysis on temperature tendencies and warm and cold extremes in Italy have been performed at national scale Toreti and Desiato (2008); Simolo et al. (2014); Fioravanti et al. (2016) and more recently, at regional level Curci et al. (2021); Morlot et al. (2023). However, a comprehensive analysis of summer heat wave characteristics is lacking. This study is focused on the analysis of several climate indicators related to summer heat extremes and heat waves on the whole Italian territory, in order to compose a quantitative picture of their different aspects in the recent period: number, frequency, duration and severity. The dataset is based upon a set of quality-controlled and homogenized daily maximum and minimum temperature data from a wide set of weather stations in Italy, covering a period of 30 years: 1991–2020. This study can benefit of a large availability of weather information over the recent past and therefore can give a precise picture of the change that the whole present Italian climate is experiencing.

Section 2 is devoted to the description of this dataset, as well as the indices calculated to evaluate the heat extremes features. Such features have been addressed by means of two separate analyses, reported in sections 3 and 4. The first one consists of a per-station analysis, which includes the extrapolation of a trend on several temperature extremes-related indices. The second one is in fact an aggregated analysis, which aims at finding a common signal related to heat waves tendencies over the whole country. The final Section 5 reports the conclusions of the present work.

2 Data and methods

2.1 Data

Daily maximum (T_{MAX}) and minimum (T_{MIN}) temperature records used in this work cover the most recent climatological reference 30-year period 1991–2020 and have been extracted from SCIA, the Italian “National system for climate data collection, processing, and dissemination” Desiato et al. (2011). SCIA collects data from different sources, including the synoptic network of the Italian Air Force Meteorological Service and several meteorological, agrometeorological, and hydrographic networks. It provides multiple quality-controlled climate statistics and indicators at different time resolutions, as well as several climatic products, through a dedicated website (scia.isprambiente.it). Data used for the processing of different climate indicators have undergone different checks, as quality assurance is the primary requirement for climate data. A first set of quality controls is applied by each data provider. Furthermore, daily data are subjected every year to a series of automated routines derived from those implemented for the Global Historical Climatological Network - GHCN Durre et al. (2010) before being processed, with the aim of detecting and invalidating erroneous data Fioravanti et al. (2016).

Homogeneity of time series is another crucial aspect for the analysis of climate variability and change, as non-climatic factors can greatly affect observed data, leading to erroneous evaluations (2020). For the purpose of providing a long-term, high-quality and homogenized dataset, T_{MAX} and T_{MIN} daily series covering the period 1961-2020 were selected according to specific data completeness and continuity criteria. Finally, they were homogenized using the `Climatol` R package (2019), following the procedure described in Fioravanti et al. (2019). This same dataset has been the basis for several Italian climate products and studies including the yearly report on climate (2023), 1991-2020 climate normals Fioravanti et al. (2022) and interpolation of relevant climate variables using the INLA-SPDE approach Fioravanti et al. (2023).

As the main focus of the present work is to investigate recent heat wave features and changes, the analysis is focused on the most recent climatological reference period, 1991-2020, which also assures the highest number of available stations. For each year, the May to September extended summer is considered, when heat waves are more intense and their potential impact on human health is stronger. As missing data make a critical element for investigation on climatic extremes, temperature series are then selected using the following criteria: i) 3 days at most can be missing in a month to make it valid; ii) each of the 5 months has to be valid to make the year valid; iii) time series with less than 70% of valid years from 1991 to 2020 are discarded. The dataset used in this work consists of 278 stations across the whole Italian territory. The spatial distribution ensures a good representativeness of the complexity of the Italian climate (Fig. 1), including stations ranging from the sea level to ~ 2300 m of altitude, with 64% of the stations below 500 m and 23% between 500 m and 1000 m.

2.2 Method

The first step of the analysis consists of calculating the number of hot days (HD) for daily T_{MAX} and T_{MIN} , for each station. A hot day is defined as a day having T_{MAX} (T_{MIN}) above the calendar day 90th percentile of the reference 1991-2020 distribution. The 90th percentile is calculated for each calendar day using a 15-day running window around it (-7 d, +7 d), with the requirement of having no less than 70% of valid data in the window. In this work, heat waves are studied among their different aspects, in terms of frequency, duration and intensity, following the ET-SCI (Expert Team on Sector-specific Climate Indices) definitions The Expert Team on Sector-Specific Climate Indices (ET-SCI) (2016). Therefore, a heat wave (HW) is hereby defined as a sequence of at least three consecutive hot days. Five aspects of HWs are considered for both the metrics T_{MAX} and T_{MIN} (Tab. 1): the number of HW events (HWN), HW frequency (HWF), duration (HWD), magnitude (HWM) and amplitude (HWA). The usage of such HW indices is well established in literature Perkins and Alexander (2013).

Furthermore, following the work in Perkins-Kirkpatrick and Lewis (2020), the cumulative heat index (ΔT_{90p}) is considered too, defined as the sum of the temperature difference with respect to the relative 90th percentile reference, across all HW days within the year. This index describes the extra heat produced by heat waves over the extended summer and can be useful for impact evaluation, giving a measure of excess heat exposure at the local level.

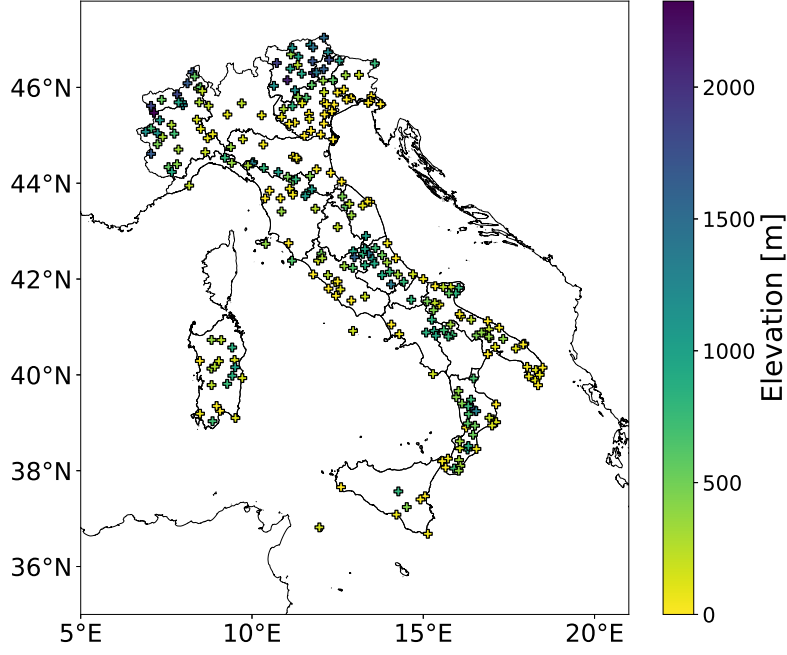


Fig. 1 Map of the weather stations selected for the study. Colors indicate the station elevation. Italian regional borders are also displayed

Table 1 Heat wave T_{MAX} and T_{MIN} -based indices: description and units

Index	description	Unit
HWN	number of individual heat waves	events
HWF	number of days contributing to heat waves	days
HWD	total duration of the longest heat wave	days
HWM	mean temperature of all heat waves	°C
HWA	max daily value in the heat wave with the highest HWM	°C
ΔT_{90p}	linear sum of temperature difference with respect to the 90 th percentile	°C

Considering both T_{MAX} and T_{MIN} , two different kinds of investigations are performed: the first one at the individual station level, the second one considering the overall behaviour of the series.

The first analysis is devoted to the extrapolation of significant trends for HD and HW-related indices for each station over the Italian territory. To evaluate the trend over the 30-year time series, the presence of at least 70% of valid data is required. A non-parametric approach is applied, using the Sen's linear trend estimator [Sen \(1968\)](#) and the Mann-Kendall test [Mann \(1945\)](#); [Kendall \(1975\)](#), a robust statistical method widely applied to study climate and hydrological time series [Lettenmaier et al. \(1994\)](#); [Donat et al. \(2013\)](#); [Panda et al. \(2017\)](#). In order to define a statistically significant

result, a threshold of 0.05 is set, which is equivalent to the request of the null hypothesis probability to be below 5%.

The second analysis, complementary to the first one, is focused on the search for a global signal related to heat waves across the Italian territory, as well as its behaviour as a function of time. In order to obtain a statistically robust result, the time series have been grouped into three subsamples, corresponding to the three decades that form the time interval analysed: 1991-2000, 2001-2010, 2011-2020. For each station, the HD and HW indices are computed per-year. The single-station distributions are then merged across all stations, within the same decade. As a result, one cumulative distribution per decade is formed and can be interpreted as a Probability Density Function (PDF). The indices HWM and HWA undergo a standardisation procedure, described at Sec. 4.1, prior to the merging. Finally, a fit is performed on each PDF, to get its overall properties. The fit functions used differ as a function of the index. Regarding HD, HWN, HWF, HWD and ΔT_{90p} , the Gamma distribution is adopted, since it proves to perform well on distributions bounded at one end and skewed. The indices HWM and HWA are fitted using a Generalised Extreme Value distribution. A similar procedure is used in [Della-Marta et al. \(2007\)](#). The fit procedure has been finalised by means of the Python LMFIT package [Newville et al. \(2015\)](#).

3 Trend analysis

Concerning the trend analysis on single stations, a subgroup of the indices only is reported. The indices chosen (HD, HWN, HWF) do not require the presence of a heat wave to be computed and can therefore benefit from the full statistics. The other indices are in fact defined after the existence of a heat wave, limiting the number of days which are valid on single stations. The requirements specified at Sec. 2.2, about the number of valid days necessary for the computation, make quite challenging the extrapolation of a trend, resulting in few significant outcomes only, over the whole dataset. This is why the authors prefer to report the indices which have a widespread solid result only.

A summary of the trends behaviour is reported in Figure 2. Overall, it can be seen a tendency toward an increase of hot days, as well as heat wave based indices, for both T_{MAX} and T_{MIN} , since the majority of results consists of positive values. Considering only statistically significant results, positive trends cover about 95% of the output, across all indices. The fraction of significant results oscillates between 35% and 46%, depending on the index. HD highlights the highest fractions, with 42% for T_{MAX} and 46% for T_{MIN} . HWN results in 35% of significant results for both T_{MAX} and T_{MIN} , while HWF features a significance of 35% for T_{MAX} and 38% for T_{MIN} .

Figures 3, 4, 5 report the spatial distribution of statistically significant trends for each index, showing the outcome of the fit over the stations, in terms of days per decade (HD, HWF) or events per decade (HWN). Positive trends are divided into three classes, while negative trends are aggregated into one class.

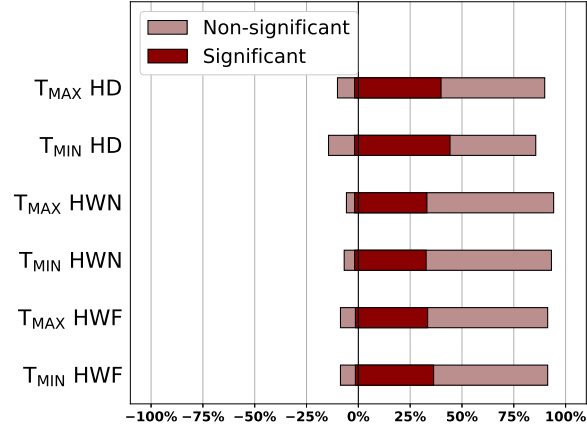


Fig. 2 Fraction of positive and negative trends obtained, reported for each index. The positive fraction is shown towards right, while the negative fraction is shown towards left. Light red: non statistically significant results; dark red: statistically significant results

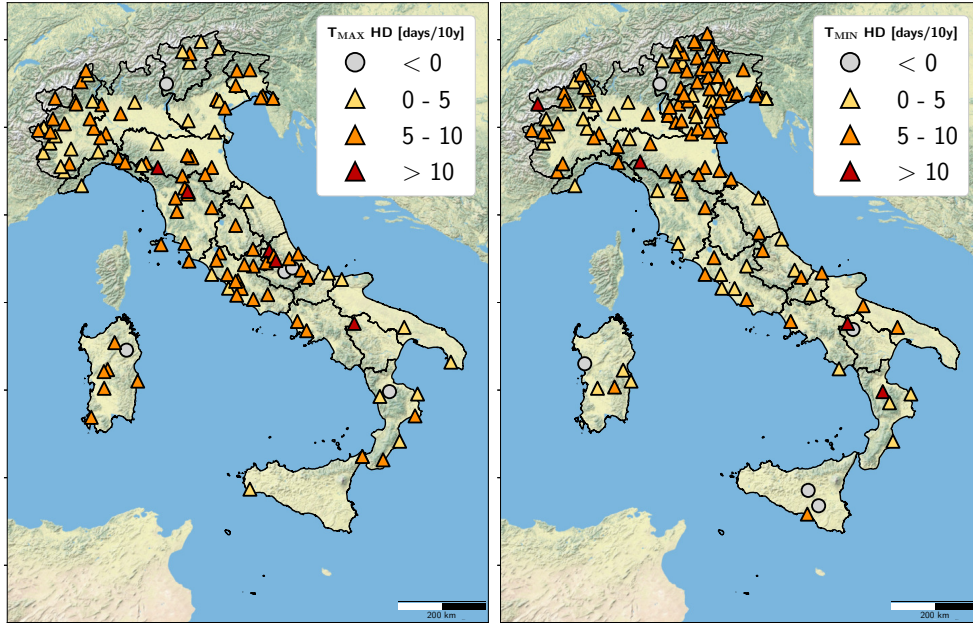


Fig. 3 Trend of T_{MAX} HD (left) and T_{MIN} HD (right) over the selected stations, in terms of days per 10 years. Reported stations have statistically significant results. Italian regional borders are also displayed

The behaviour of the three indices consists of a clear and widespread positive trend across the whole country. Still, the effect has a greater prominence in some areas with respect to others. Regarding T_{MAX} , a cluster of significant positive results can be

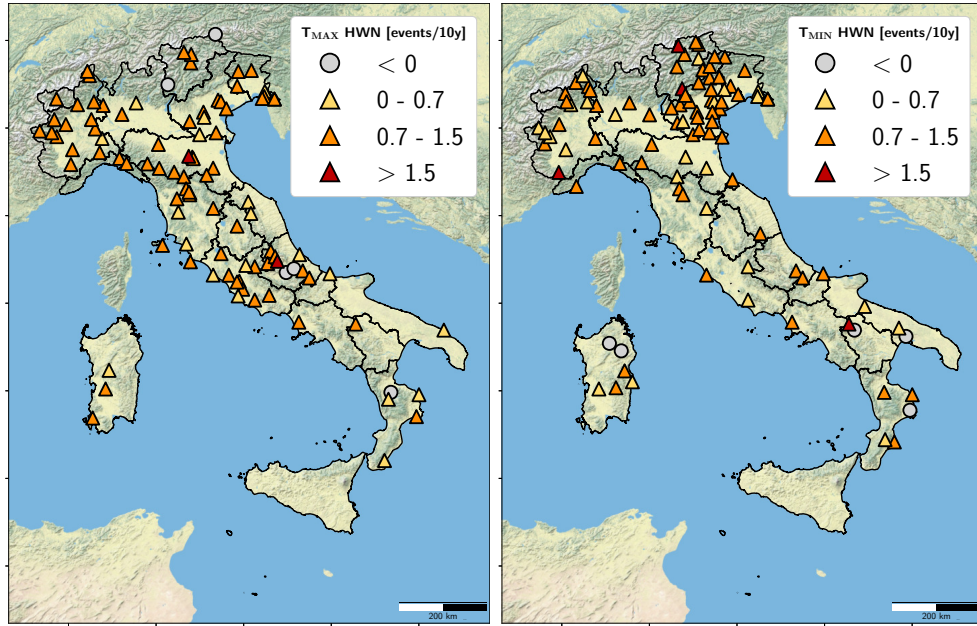


Fig. 4 Trend of T_{MAX} HWN (left) and T_{MIN} HWN (right) over the selected stations, in terms of events per 10 years. Reported stations have statistically significant results. Italian regional borders are also displayed

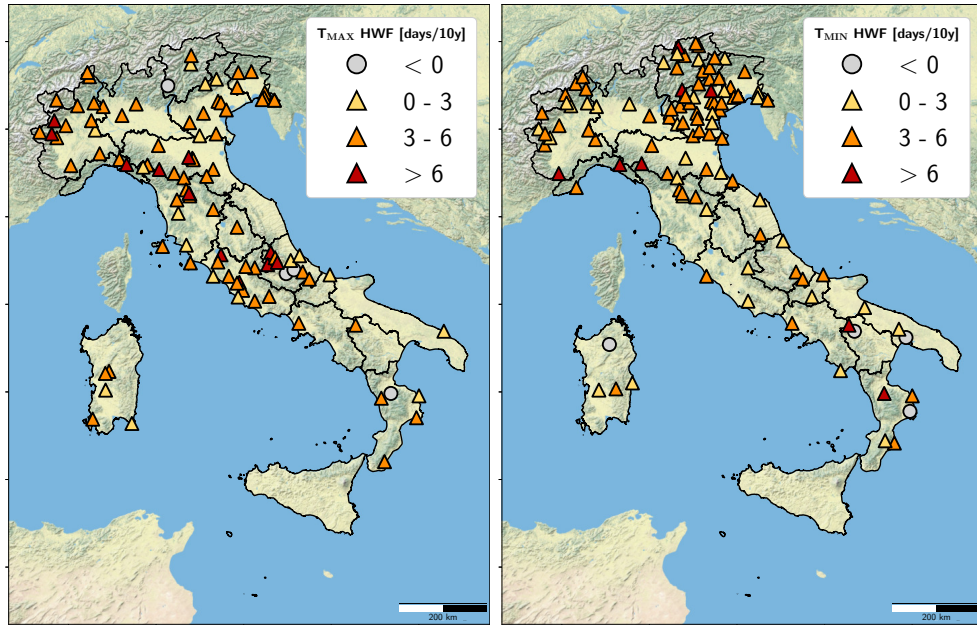


Fig. 5 Trend of T_{MAX} HWF (left) and T_{MIN} HWF (right) over the selected stations, in terms of days per 10 years. Reported stations have statistically significant results. Italian regional borders are also displayed

identified in Central Italy, from the Apennines down to the Tyrrhenian Sea. The T_{MIN} -related results have instead a significant cluster in the North-East. Southern regions collect on average fewer significant results with respect to the rest of the country, although still largely positive.

Quantitatively, the number of hot days show an increase higher than 5 days per decade for 74% (T_{MAX}) and 68% (T_{MIN}) of the displayed stations (Fig. 3). The trend values obtained for the HWN and HWF indices describe a significant increase of heat waves presence in Italy too. However, the magnitude for HWN (Fig. 4) is not numerically high, being around 1 in most cases, suggesting that the total number of heat waves per year has only slightly increased over the 30-years period, for both T_{MAX} and T_{MIN} . On the other hand, the clear positive HWF trend indicates longer periods of extreme heat, with 77% of displayed stations for T_{MAX} which have already experienced an increase of more than 3 days of HW per decade (Fig. 5). As a result, despite the total number of HW per year has just weakly changed, their persistency has significantly increased. The signal appears slightly less strong for T_{MIN} HWF, with 63% of the stations having an increase per decade larger than 3 days.

4 Decennial analysis

The global evolution of hot days and heat waves on national scale is investigated by analysing decennial changes for the whole set of series, over three 10-year periods (1991-2000, 2001-2010 and 2011-2020), according to the methodology reported in 2.2. A standardisation procedure is applied to the heat wave magnitude (HWM) and amplitude (HWA) indices, as described in the following section.

4.1 Standardisation

Although a HW is here defined with respect to the 90th percentile of the reference distribution, which is specific of each station, the indices HWM and HWA consist of an absolute temperature, either as a mean or a peak value. If the single-station distributions were merged as they are to form the decade PDF, temperature ranges far away from each other (from the Alps to African latitudes) would be combined together and the resulting distribution would be meaningless.

In order to avoid the issue, as described in Della-Marta et al. (2007), each station distribution, for HWM and HWA, is standardised by its mean and standard deviation, as in the following:

$$X'_i = \frac{(X_i - \bar{X})}{\sigma_X}, \quad (1)$$

where X_i is the original series, $\bar{X}$ is the series mean, σ_X its standard deviation and X'_i denotes the standardised series. After the procedure, the mean of all series belonging to the same decade, as well as the standard deviation, are added back to the PDF, to quantify changes across the decades. For clarity, indices which are not subject to the standardisation procedure are henceforth labeled as "Group 1", while HWM and HWA as "Group 2".

4.2 Fit results and uncertainty

The output of the fit is considered both in terms of the functions shape and the fitted parameters. Figures 6 – 12 show a comparison of the fitted functions for all the indices considered, separated per decade.

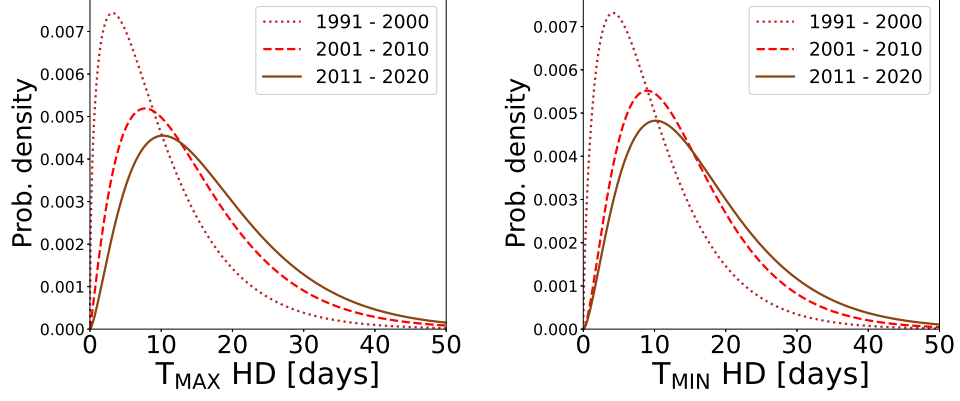


Fig. 6 HD PDF for the three decades. Left: daily T_{MAX} ; right: daily T_{MIN}

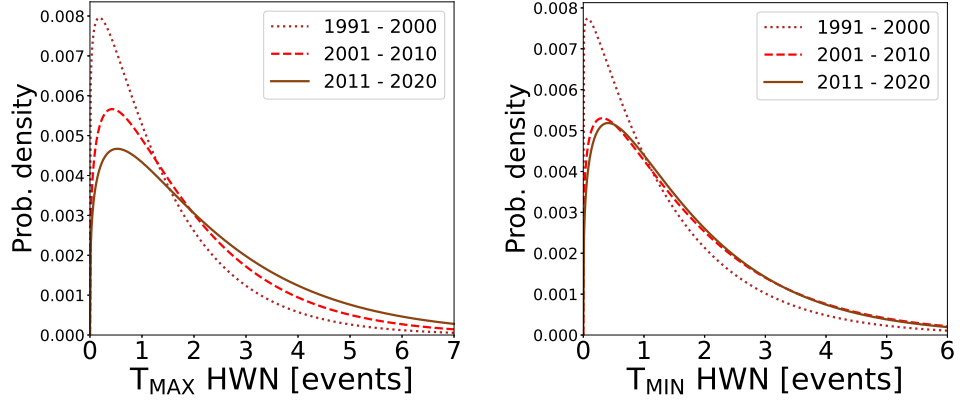


Fig. 7 HWN PDF for the three decades. Left: daily T_{MAX} ; right: daily T_{MIN}

In most cases, the PDF shapes have a tendency to move towards high values, enhancing the weight of the right tail. A more populated tail means a higher probability of extreme heat events occurrence. For some indices, the effect is more visible than for others. Regarding Group 2, for instance, it's harder to extrapolate a tendency by looking at the shapes only.

In order to quantify the differences between decades, some representative parameters

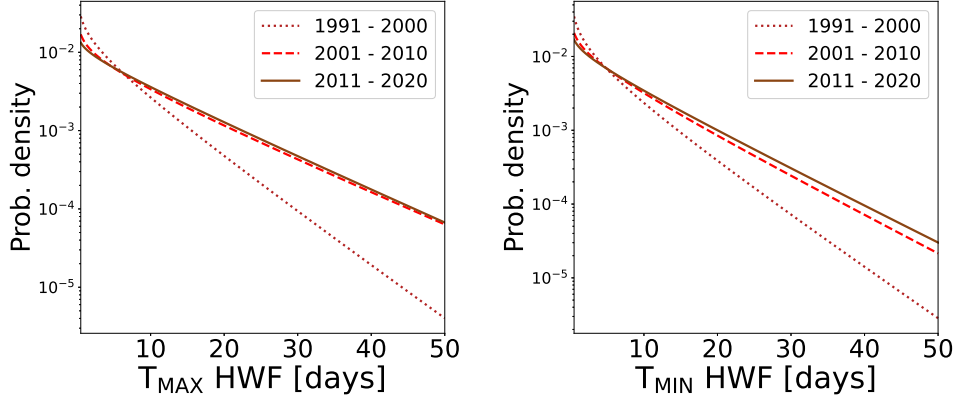


Fig. 8 HWF PDF for the three decades. Left: daily T_{MAX} ; right: daily T_{MIN}

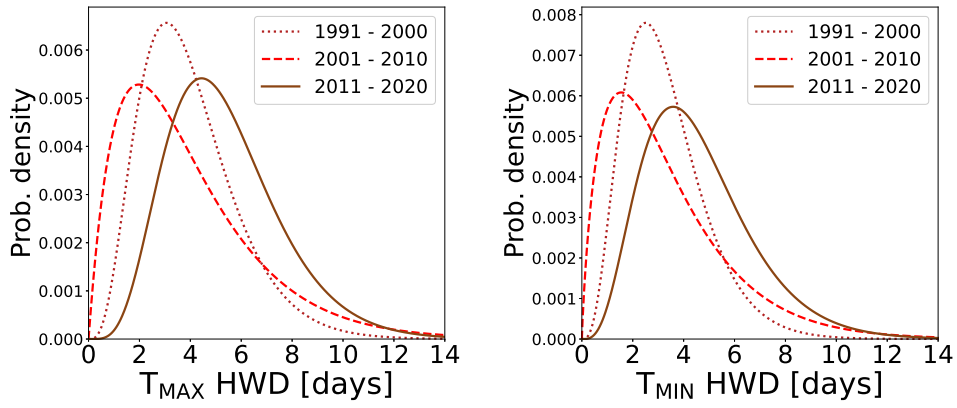


Fig. 9 HWF PDF for the three decades. Left: daily T_{MAX} ; right: daily T_{MIN}

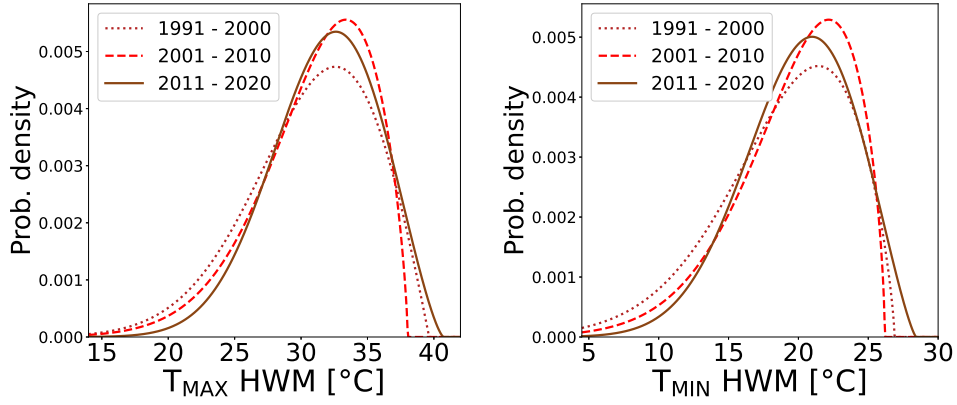


Fig. 10 HWF PDF for the three decades. Left: daily T_{MAX} ; right: daily T_{MIN}

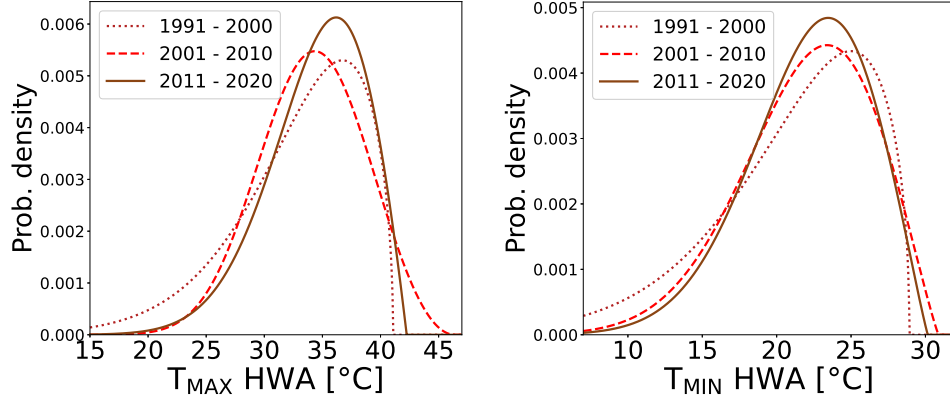


Fig. 11 HWA PDF for the three decades. Left: daily T_{MAX} ; right: daily T_{MIN}

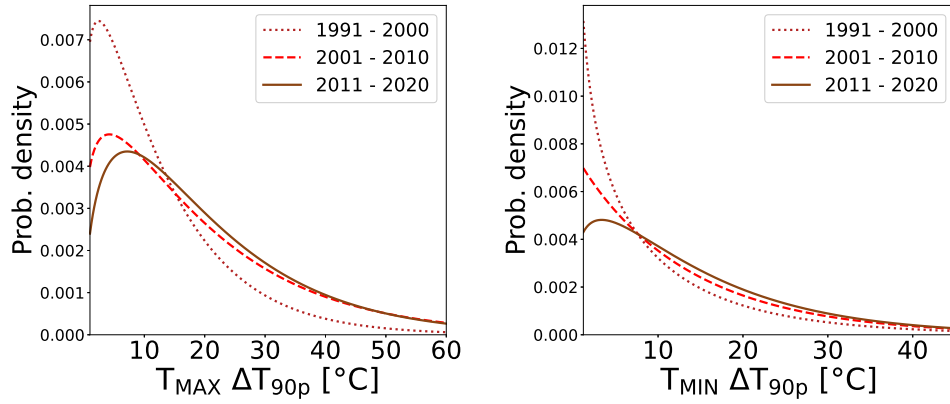


Fig. 12 ΔT_{90p} PDF for the three decades. Left: daily T_{MAX} ; right: daily T_{MIN}

are taken into account. Such parameters depend upon the fit function used. For Group 1, the mean and standard deviation of the Gamma distribution can be conveniently expressed in terms of the shape parameter ξ and of the scale parameter σ [Thon \(1958\)](#), as:

$$\begin{aligned} \text{Mean} &= \xi\sigma \\ \text{Std Dev.} &= \sqrt{\xi\sigma^2}, \end{aligned} \tag{2}$$

taking the standard deviation as the square root of the variance.

For the Group 2 indices, fitted using a Generalised Extreme Value distribution, the location parameter μ and the scale parameter σ are considered. μ and σ are related to the overall position and spread of the function, respectively. Therefore, they can be used as probes when looking for changes across decades. An increasing μ implies that

the distribution is overall shifting towards higher values, while a high σ reflects a distribution less concentrated around the mean and more prone to pick values in the tails. The behaviour of the parameters mentioned above is then studied as a function of the decade. Each parameter has an associated statistical uncertainty, either computed directly during the fitting procedure (ξ , μ , σ) or through error propagation (mean, std. deviation). However, the final fitted values may be also affected by biases introduced by the fitting procedure itself. The choice of the PDF distributions binning, in particular, is seen to affect the final result to a non negligible extent, for some of the indices. In order to give a quantitative assessment of the binning impact, large variations of the number of bins are applied and for each configuration a separate fit is performed. For each parameter, the standard deviation of the results distribution is taken as systematic uncertainty, which is added in square sum to the statistical term to form the total uncertainty.

Figures 13, 14, 15, 16 report the resulting parameters with the associated uncertainties, including the contribution from the binning, as a function of the decade. Most parameters show a growing tendency with time, with few exceptions.

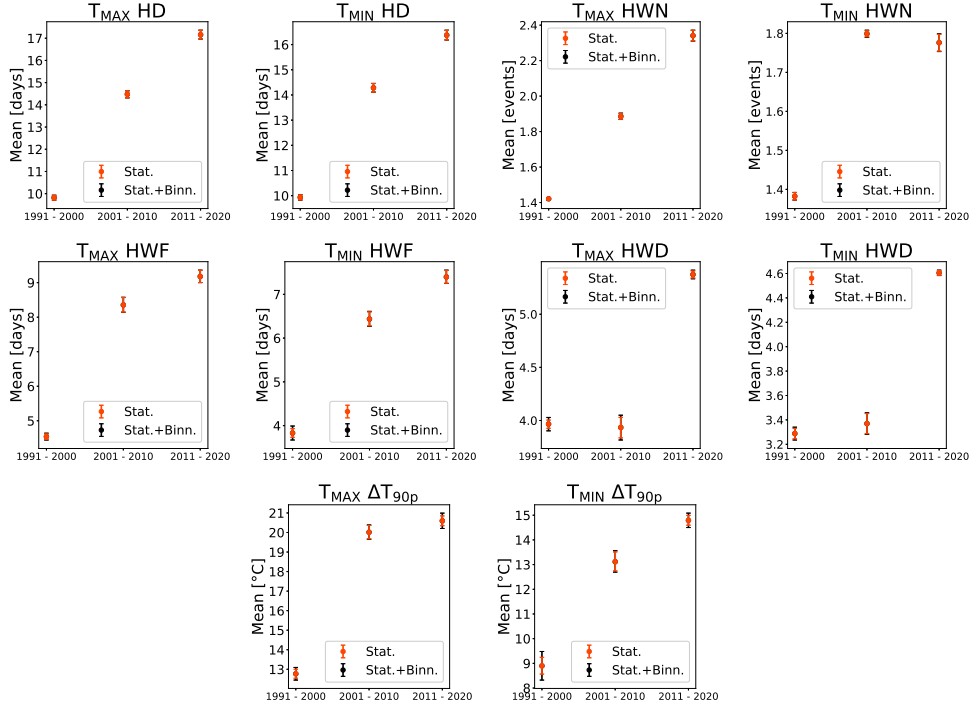


Fig. 13 Fitted mean for Group 1 indices, as a function of the decade. Both statistical-only (red) and total (black) uncertainties are displayed

For some parameters, extrapolating a linear trend appears straightforward, as values are aligned with little deviations. For Fig. 13, this is the case of HD, HWF, T_{MAX}

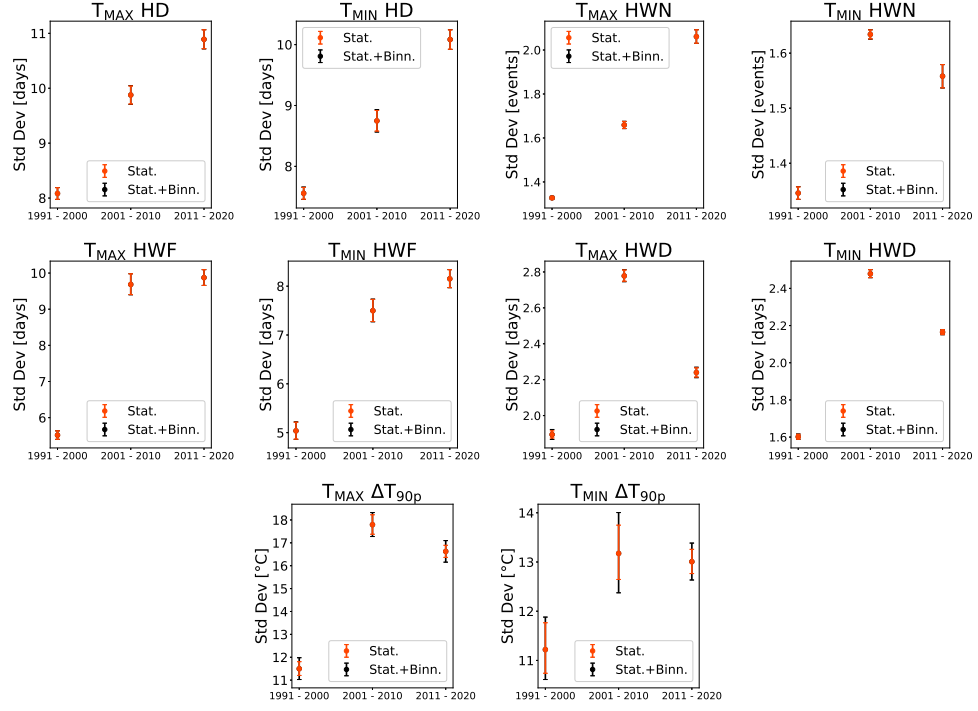


Fig. 14 Fitted standard deviation for Group 1 indices, as a function of the decade. Both statistical-only (red) and total (black) uncertainties are displayed

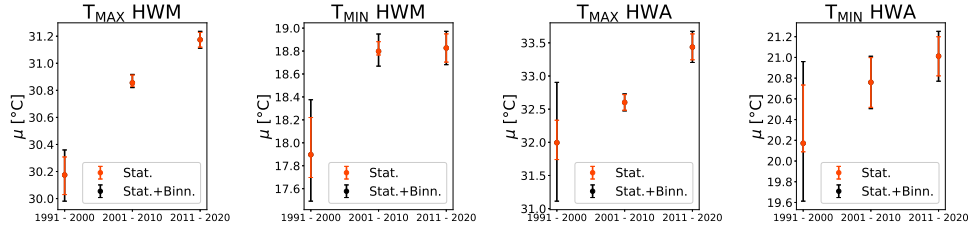


Fig. 15 Fitted μ for Group 2 indices, as a function of the decade. Both statistical-only (red) and total (black) uncertainties are displayed

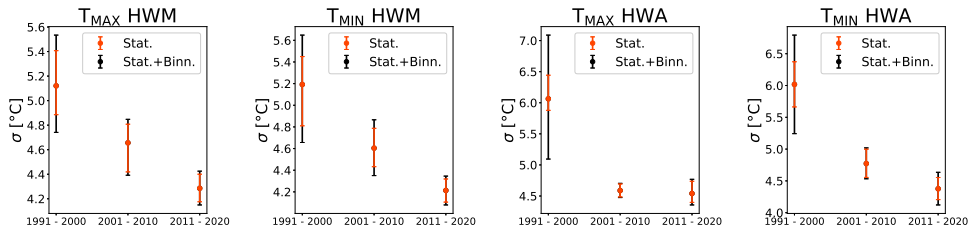


Fig. 16 Fitted σ for Group 2 indices, as a function of the decade. Both statistical-only (red) and total (black) uncertainties are displayed

HWN and $T_{\text{MIN}} \Delta T_{90p}$. For Fig. 14, a similar pattern can be seen for HD, T_{MAX} HWN and T_{MIN} HWF. For Fig. 15, all parameters behave quite similarly, except for T_{MIN} HWM. A different trend, although consistent among parameters, can be observed in Fig. 16, where values are decreasing as a function of time. Generally, figures 13, 14, 15 depict a climatic scenario where the presence of hot days and heat waves in Italy increases, with an increasing probability of experiencing extreme conditions at the tail of the distributions. The simultaneous growing values for HD, HWN and HWF support the results reported in Sec. 3, confirming longer periods of extreme heat as a function of time. The HWD and ΔT_{90p} results also strengthen the picture of more persistent and hotter heat waves. Trends observed in Fig. 15 follow the same direction. It can be noted that values for the 2001–2010 decade sometimes lie above the 2011–2020 ones, a feature that can be related to the record-breaking 2003 summer [García-Herrera et al. \(2010\)](#). However, none of the parameters reported in the figures 13, 14 and 15 shows a value for the 2011–2020 decade which is lower than the 1991–2000 one.

The interpretation of Fig. 16 is in fact less obvious. The behaviour of μ for HWA and HWM tends to support a picture with increasing temperatures within heat waves (Fig. 15), however their PDFs seems to shrink around the mean (Fig. 16). A possible explanation is that the σ decrease is due to the decrease of the area of the left-side tail of the PDFs, visible in figures 10 and 11. According to this interpretation, temperatures recorded during heat wave events are concentrating at high values, excluding low ones. Regarding the evolution of the PDFs over time, the study of [Simolo et al. \(2014\)](#) already found a significant enhancement of the mean value in the distribution of warm days until 2007. However, the same study found no evidence of a modification in the shape of the function, indicating that such change should have occurred only recently. In order to provide an estimate of the parameters tendency, a separate linear regression model is applied on each parameter displayed in figures 13, 14, 15, 16. The fit is still performed by means of the Python LMFIT package [Newville et al. \(2015\)](#). Results are summarised in tables 2 and 3, where the slope parameters are reported for Group 1 and Group 2 indices, respectively.

Table 2 Resulting slope from linear regression, for Group 1 indices. The associated statistical uncertainty is also reported. Upper section: T_{MAX} -based indices; lower section: T_{MIN} -based indices. The asterisk marks results whose value is above twice the uncertainty

T_{MAX}	HD [d/10y]	HWN [ev./10y]	HWF [d/10y]	HWD [d/10y]	ΔT_{90p} [°C/10y]
Mean	3.84±0.54*	0.46±0.01*	2.46±0.60*	0.73±0.22*	4.12±1.87*
Std dev.	1.45±0.19*	0.35±0.02*	2.33±0.74*	0.18±0.35	2.56±1.99
T_{MIN}	HD [d/10y]	HWN [ev./10y]	HWF [d/10y]	HWD [d/10y]	ΔT_{90p} [°C/10y]
Mean	3.39±0.58*	0.29±0.15	1.77±0.43*	0.70±0.17*	2.69±0.70*
Std dev.	1.26±0.03*	0.17±0.13	1.56±0.43*	0.28±0.24	0.82±0.41

The estimated trends are a confirmation of the results discussed above and reflect the tendencies observed in figures 13, 14, 15 and 16. With respect to Group 1, a solid

Table 3 Resulting slope from linear regression, for Group 2 indices. The associated statistical uncertainty is also reported. Upper section: T_{MAX} -based indices; lower section: T_{MIN} -based indices. The asterisk marks results whose value is above twice the uncertainty

T_{MAX}	HWM [$^{\circ}\text{C}/10\text{y}$]	HWA [$^{\circ}\text{C}/10\text{y}$]
μ	$0.39 \pm 0.11^*$	$0.81 \pm 0.06^*$
σ	$-0.40 \pm 0.03^*$	-0.14 ± 0.33
T_{MIN}	HWM [$^{\circ}\text{C}/10\text{y}$]	HWA [$^{\circ}\text{C}/10\text{y}$]
μ	0.24 ± 0.28	$0.35 \pm 0.11^*$
σ	$-0.45 \pm 0.05^*$	$-0.57 \pm 0.26^*$

positive linear trend is obtained in the mean for all indices except for T_{MIN} HWN, as well as in the standard deviation for most of the indices. For Group 2 indices, T_{MAX} HWM, T_{MAX} HWA and T_{MIN} HWA show a clear increasing linear trend of the location parameter μ . A negative trend is confirmed for the scale parameter σ of HWM and HWA PDFs, more prominently for T_{MAX} than T_{MIN} . Results for HWF are numerically consistent with those found for Italy by Perkins-Kirkpatrick and Lewis (2020) and Rousi et al. (2022), although the latter study uses satellite data and both studies extrapolate trends on a longer time scale. It must be taken into account, anyway, that both these two works are wide-range analyses (global or pan-European) and are based on reanalysis data. Moreover, Rousi et al. (2022) considers only the July-August period. Results compatible with Perkins-Kirkpatrick and Lewis (2020) are obtained for HWD too, while slightly different values are found for HWM and ΔT_{90p} , but not too far considering the full uncertainty. Close results for ΔT_{90p} are found instead when comparing with Rousi et al. (2022). In any case, it's worth to mention that this work is specifically targeted to the Italian territory, while comparisons with Perkins-Kirkpatrick and Lewis (2020) and Rousi et al. (2022) are performed by looking at the grid points falling into the same area.

Being the result of the combination of all available stations, the trends reported in tables 2 and 3 represent an average effect over the whole country and do not take into account local extremes. They give, however, a global picture of high temperature extremes, which are undoubtedly moving towards higher and higher occurrence probability.

5 Conclusions

The present study provides a comprehensive assessment of heat extremes over the whole Italian area, getting benefit of all the available statistics regarding in-situ measurements. To give a quantitative picture which is close to the current population experience and to employ the highest amount of temperature data, the analysis period

considered ranges from 1991 to 2020, very close to present days. An updated quality controlled and homogenised dataset of daily temperature series was examined to address heat extremes and in particular heat waves. In order to provide a common metric for the global climate community, a set of well-established indices is adopted for the characterization of the phenomenon.

The present study is developed in two different branches. One focused on local features of heat extremes and one which looks at their behaviour as a whole, as the double level of the information (global and local) can provide fruitful insights for trends comparison. Both analyses agree in describing a reinforcement of hot days and heat waves over the country, during the 30 years just passed.

At the individual station level, the trend analysis highlights a tendency toward an increase of hot days, as well as heat wave number and frequency, for both minimum and maximum temperature-based indices. In general, most of trends consists of positive values (95% among the significant ones), with the fraction of statistically significant results oscillating between 35% and 46%, depending on the index. The number of hot days show an evident increase, with trend values higher than 5 days per decade for 74% (T_{MAX}) and 68% (T_{MIN}) of the stations. While the total number of heat waves per year has only slightly increased over the 1991-2020 period, for both T_{MAX} and T_{MIN} , the heat wave frequency shows a clearer trend. The results indicates that 77% of stations for T_{MAX} and 63% of the stations for T_{MIN} have registered an increase of more than 3 days of heat wave per decade, which translates into ~ 10 days over the whole period. The trend maps present a warming tendency over the whole national territory. Furthermore, the trend spatial pattern indicates the areas where the effects appear stronger, like the central regions for T_{MAX} and the North-East for T_{MIN} .

The PDFs of hot days and heat waves related indices highlight the global behaviour of Italian heat extremes, by comparing 1991-2000, 2001-2010 and 2011-2020 decades. For most indices, a shift from 1991-2000 to the last decade is observed, resulting into an increasing occurrence probability for extreme heat conditions on the right side of the distributions. Although some differences across the indices are present, the PDFs confirm the global picture where heat extremes are increasing their presence over Italy in terms of persistence and intensity, i.e. temperatures that the population is exposed to. The total number of hot days, for instance, increases on national scale by a rate of 3–4 days per decade. The magnitude and amplitude indices show a more pronounced trend for T_{MAX} ($+0.39^{\circ}\text{C}/\text{decade}$ and $+0.81^{\circ}\text{C}/\text{decade}$, respectively) than for T_{MIN} . Similarly, a clear positive tendency of the cumulative heat index is observed, stronger for T_{MAX} ($+4.1^{\circ}\text{C}/\text{decade}$) than for T_{MIN} ($+2.7^{\circ}\text{C}/\text{decade}$).

All the results presented in this work contribute to compose an alarming picture for the Italian climate, experiencing the effects of the atmosphere warming at worrisome rates. The outcomes of this work, although based on a relatively short period of 30 years, are statistically very strong and could represent a solid base when planning actions to protect the most vulnerable sectors of the population from heat damages.

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Competing Interests

The authors have no competing interests to declare that are relevant to the content of this article.

Author Contributions

All authors contributed to the study conception and design. Material preparation was performed by Emanuela Piervitali and Giulio Settanta. Data collection and processing were performed by Piero Frascchetti, Francesca Lena, Walter Perconti, Emanuela Piervitali and Giulio Settanta. The analysis was performed by Giulio Settanta. The first draft of the manuscript was written by Giulio Settanta and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Data Availability

The dataset used in the current study has been selected from the SCIA-ISPRA system. Most of it is publicly available and can be accessed at www.scia.isprambiente.it. In respect of the policy of each data provider, some data are not accessible through the SCIA website and are either available through the regional websites or should be requested directly to the data owners.

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